Aymane El Firdoussi

Website: elfirdoussilab1.github.io
Email: aymane.elfirdoussi@telecom-paris.fr
GitHub: https://github.com/elfirdoussilab1

Education

Institut Polytechnique de Paris — Palaiseau, France
M.Sc. in Mathemtics, Vision and Learning (MVA) - GPA: 4.0/4.0 — Sep 2024–Oct 2025

- Pursued the **MVA** (Mathématiques, Vision et Apprentissage) master's program.
- **Thesis:** Effective Fine-tuning via Base Model Rescaling: A Random Matrix Theory Perspective.
  Advisors: Prof. Alexandre d'Aspremont (ENS Paris) & Prof. Martin Jaggi (EPFL).
- **Coursework:** Deep Learning, Reinforcement Learning, Large Language Models, Graphs in ML, Generative Modeling, Object Recognition and Computer Vision, Random Matrix Theory.

Télécom Paris — Palaiseau, France
M.Sc in Engineering: Applied Maths and Computer Science - GPA: 4.0/4.0 — Sep 2021–Dec 2025

- Pursued a competitive research track under the supervision of Prof. **Pascal Bianchi.**
- **Coursework:** Statistics, Probability Theory, Machine Learning, Algorithms and data structures, Information Theory, Signal Processing.

Lycée Ibn Timiya — Marrakesh, Morocco
Preparatory classes in Mathematics, Physics and Computer Science — Sep 2019–Aug 2021

- **Coursework:** Real and Complex Analysis, General topology, General and Linear Algebra.

Experience

Google DeepMind — Paris, France
Research intern — Nov 2025 - Mai 2026

- **Supervisors:** Dr. Romuald Elie & Dr. Arnaud Doucet.
- Working, within the **Frontier AI** unit, on fine-tuning Diffusion Models with Reinforcement Learning techniques (especially policy gradient methods).

EPFL, Machine Learning and Optimization lab — Lausanne, Switzerland
Visiting student (Master thesis) — April 2025 - Oct 2025

- **Supervisors:** El Mahdi Chayti & Prof. Martin Jaggi.
- Worked on improving the generalization capabilities of reparameterization-based fine-tuning methods (such as LoRA). The full master thesis report can be found [here].
- Proposed α-LoRA, a new variant of Low Rank Adaptation algorithm for fine-tuning LLMs, which provably improves the generalization capabilities of the fine-tuned model. The theoretical efficiency of this algorithm was proved with Random Matrix Theory tools. This work [1] is under review.

Technology Innovation Institute (TII) — Abu Dhabi, UAE
Research intern — Mar 2024 - Sep 2024

- **Supervisor:** Dr. Mohamed El Amine Seddik
- Characterized the impact of training models on a mixture of real and synthetic data, and showed that the distribution shift between these two types of data determines the quality of synthetic samples. Our work gave more insights into why Model Collapse happens for Large Language Models. Work [2] published at **ICLR 2025**.

– Proposed a robust classification loss resilient to label noise in the high-dimensional regime, with theoretical guarantees thanks to Random Matrix Theory. Work [3] published at HiLD workshop at **ICML 2024**.

EPFL, Distributed Computing Lab Lausanne, Switzerland
Visiting student (research internship) July 2023 - Jan 2024

– **Supervisors:** Dr. Youssef Allouah & Prof. Rachid Guerraoui
– Worked on improving the robustness of Federated Learning algorithms being faced to attacks by Byzantine agents.
– Contributed to the design and implementation of DECOR, a novel state-of-the-art Decentralized Learning algorithm with optimal performance and Differential Privacy guarantees (privacy guarantees against data leakage). Work [4] published at **ICML 2024**.

Publications

[1] **α-LoRA: Effective Fine-Tuning via Base Model Rescaling**, **Aymane El Firdoussi**, El Mahdi Chayti, Mohamed El Amine Seddik, Martin Jaggi, *preprint under review.*

[2] **Maximizing the Potential of Synthetic Data: Insights from Random Matrix Theory**, **Aymane El Firdoussi**, Mohamed El Amine Seddik, Soufiane Hayou, Reda Alami, Ahmed Alzubaidi, Hakim Hacid, *International Conference on Learning Representations (ICLR) 2025.*

[3] **High-dimensional Learning with Noisy Labels**, **Aymane El Firdoussi**, Mohamed El Amine Seddik, *HiLD workshop, International Conference on Machine Learning (ICML) 2024.*

[4] **The privacy power of correlated noise in decentralized learning**, Youssef Allouah, Anastasia Koloskova, **Aymane El Firdoussi**, Martin Jaggi, Rachid Guerraoui, *International Conference on Machine Learning (ICML) 2024.*

Selected Projects

Composed Image Retrieval [report][code] Dec 2024 - Jan 2025

– Evaluated the impact of modifying the multimodal embeddings of the BLIP model on composed image retrieval tasks.
– Showed that BLIP's multimodal embedding layer does not fully capture all the necessary information from both modalities, and proposed an adaption to this latter.

Stein Variational Gradient Descent [Litterature review][Final report] Dec 2024 - Jan 2025

– Project conducted during my research assistantship of Prof. Pascal Bianchi at Télécom Paris.
– Provided a theoretical analysis of the convergence of the Stein Variational Gradient Descent (SVGD), an algorithm used to sample from any target probability distribution, and introduced a new stochastic variant of this latter.

Stereogame [code] Dec 2024 - Jan 2025

– In a team of 4 students, we created a duo game for 3D TVs called Stereogame, where players can both use the full screen and play against each other at the same time. Thus, no need for screen splitting anymore!
– The game was developped with OpenGL (for graphical design) and the pygame library.

Skills

- **Programming:** Python, C++, C, SQL.
- **Frameworks:** Pytorch, JAX, NumPy.
- **Languages:** Arabic (native), French (bilingual), English (Fluent), Spanish (Beginner).

Scholarships and Awards

- **Ibn Rochd scholarship (2021-2025):** Received an excellence scholarship from Ibn Roch Foundation (FIRSI, Morocco) following my admission to one of the most prestigious Grande École (universities) in France.
- **Winner of MILA's Morocco Solidarity AI Hackathon (Sep 2023):** In the context of the AI Hackathon organized by MILA, our "DeepAster" team designed a vision model to classify the degree of damage of each area in Morocco, based on satellite images. Our solution secured the fourth place (total of 5 winning teams) of the hackathon, and we were awarded 5, 000$ AWS credits.
- **Ranked 1st in Marrakesh and 15th in Morocco in the Common National Exam (July 2021):** An exam with more than 5,000 candidates applying for the most prestigious Moroccan engineering universities.

Extracurricular Activities

- **Member of AtlasIA** 2024–Present
  - AtlasIA is a Moroccan community gathering many engineers and researchers in Machine Learning. It is committed to developing AI models that resonate with Moroccan values, identity, and culture.
  - Lead the project called: **Darija-Chatbot-Arena** where we created a platform to provide a comprehensive evaluation of LLMs' capabilities in understanding and generating responses in Darija, the Moroccan Arabic Dialect, through user feedback. A detailed analysis of the obtained results can be found in our blog post on HuggingFace: [blog]
- **Past member of Sublimaths** March 2023–June 2023
  - Sublimaths is the Mathematics club of Télécom Paris. It's core activity is organizing workshops for primary and middle school students to teach them some mathematical arguments in a fun and enjoyable way. The goal is to beautify the mathematics for young teenagers and give them a new taste of this field.
  - Organized a workshop on "Algorithms" on April 2023. More details can be found in their official website [link].